Tutorial 9: solution

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2025-04-17

Set up

As usual, we start by creating a path object and loading our packages.

```
rm(list=ls())
pacman::p_load(dplyr,ggplot2,plm,stargazer)
path = "C:/Users/mateomoglia/Dropbox/courses/polytechnique/2025_eco1s002/tutorial9"
```

Open the data

The dataset has already been cleaned. It consists on labour force participation information as well as some household-level information on workers in France. It consists only of couples.

```
# Open the dataset
raw_data = read.csv2(paste0(path,"/raw/lfs_women.csv"),sep=",")

# Create the estimation sample and create relevant variables
est_sample = raw_data %>%
  mutate(lfs = ifelse(ntravail == 1, 1, 0),
         mother = ifelse(nsexe == 2 & mne1 > 0, 1, 0),
         large_family = ifelse(mne1 > 2, 1, 0),
         full_time = ifelse(ntpp == 2,1,0),
         high_skilled = ifelse(ndiplo >= 5,1,0)) %>%
  mutate(mother_x_large_fam = mother*large_family) %>%
  mutate(iv_twins = mother*hh_twin_after,
         iv_ssex  = mother*same_sexe)
```

Model

We are interested in estimating the effect of having a large family and being a mother on labour force participation. We want also to estimate the effect of being the mother of a large family, thus the interaction term `mother_x_large_fam`. The model writes:

$$\mathbb{P}(LFS_i = 1) = \beta_0 + \beta_1 \mathbf{1}(\texttt{LargeFamily}_i = 1) + \beta_2 \mathbf{1}(\texttt{Mother}_i = 1) + \beta_3 \mathbf{1}(\texttt{LargeFamily}_i = 1) \times \mathbf{1}(\texttt{Mother}_i = 1) + e_i$$

Where we estimate the βs.

OLS estimation

We estimate the model:

```r
summary(lm(lfs ~ mother + large_family + mother_x_large_fam, data = est_sample))
```

```
##
## Call:
## lm(formula = lfs ~ mother + large_family + mother_x_large_fam,
##     data = est_sample)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.8955  0.1045  0.1321  0.2073  0.3800
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.895473   0.006553 136.643   <2e-16 ***
## mother             -0.102791   0.009383 -10.955   <2e-16 ***
## large_family       -0.027593   0.011297  -2.442   0.0146 *
## mother_x_large_fam -0.145090   0.016049  -9.041   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3798 on 9955 degrees of freedom
##   (19 observations effacées parce que manquantes)
## Multiple R-squared:  0.06057,    Adjusted R-squared:  0.06029
## F-statistic: 213.9 on 3 and 9955 DF,  p-value: < 2.2e-16
```

Coefficients are all statistically significant at the 0.1% level, except for `large_family` which is significant at the 5%. However, this regression might suffer for omitted variable bias, as other variables might jointly affect having a large family and labour force participation. We add the skill level as well as age:

```r
summary(lm(lfs ~ mother + large_family + mother_x_large_fam + high_skilled + log(nag),
           data = est_sample))
```

```
##
## Call:
## lm(formula = lfs ~ mother + large_family + mother_x_large_fam +
##     high_skilled + log(nag), data = est_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.04484  0.01739  0.13401  0.21113  0.61809
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.043063   0.077864   0.553   0.5802
## mother             -0.106630   0.009331 -11.427   <2e-16 ***
```

```
## large_family       -0.021165   0.011040  -1.917   0.0553 .
## mother_x_large_fam -0.130339   0.015687  -8.309   <2e-16 ***
## high_skilled        0.152712   0.007521  20.304   <2e-16 ***
## log(nag)            0.206542   0.020598  10.027   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3705 on 9934 degrees of freedom
##   (38 observations effacées parce que manquantes)
## Multiple R-squared:  0.1054, Adjusted R-squared:  0.105
## F-statistic: 234.2 on 5 and 9934 DF,  p-value: < 2.2e-16
```

We also add household-level fixed effect to account for any time-invariant household characteristics:

```r
summary(plm(lfs ~ mother + large_family + mother_x_large_fam + high_skilled + log(nag),
            index = c("idmen"), data = est_sample))
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = lfs ~ mother + large_family + mother_x_large_fam +
##     high_skilled + log(nag), data = est_sample, index = c("idmen"))
##
## Unbalanced Panel: n = 4989, T = 1-2, N = 9940
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -0.736029 -0.089561  0.000000  0.089561  0.736029
##
## Coefficients:
##                     Estimate Std. Error t-value  Pr(>|t|)
## mother             -0.0939148  0.0094284 -9.9608 < 2.2e-16 ***
## mother_x_large_fam -0.1285179  0.0145286 -8.8459 < 2.2e-16 ***
## high_skilled        0.0536238  0.0131455  4.0793 4.589e-05 ***
## log(nag)            0.3678675  0.0592860  6.2050 5.917e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    649
## Residual Sum of Squares: 570.74
## R-Squared:      0.12059
## Adj. R-Squared: -0.76682
## F-statistic: 169.587 on 4 and 4947 DF, p-value: < 2.22e-16
```

Because the household fixed effect is colinear with having a large family, the coefficient for large family is dropped.

IV estimation

There is endogeneity concern due to reverse causality between having a large family and labour force participation. For instance, working increases income, therefore having children is made economically viable.

Two instruments can be used. The goal of the instrument is to find an exogenous variation in family size.

- **Having twins as a second birth**. Indeed, the decision of having a second child might be endogenous, but having twins is not. It is a plausible exogenous variation in family size, going from 1 to 3, instead of 1 to 2, as planned
- **Having two first child the same gender**. Indeed, there is a preference for having a balanced gender composition. Hence, there might be a will to have a third child, potentially of a different gender, to balance the gender composition, independantly of labour force participation.

```r
summary(plm(lfs ~ mother + high_skilled + log(nag) + mother_x_large_fam  |
              iv_twins + high_skilled + mother + log(nag),
            index = c("idmen"),
            data = est_sample))
```

```
## Oneway (individual) effect Within Model
## Instrumental variable estimation
##
## Call:
## plm(formula = lfs ~ mother + high_skilled + log(nag) + mother_x_large_fam |
##     iv_twins + high_skilled + mother + log(nag), data = est_sample,
##     index = c("idmen"))
##
## Unbalanced Panel: n = 4989, T = 1-2, N = 9940
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -0.689447 -0.098517  0.000000  0.098517  0.689447
##
## Coefficients:
##                     Estimate Std. Error z-value  Pr(>|z|)
## mother             -0.180254   0.031374 -5.7453 9.177e-09 ***
## high_skilled        0.071207   0.014849  4.7954 1.623e-06 ***
## log(nag)            0.425703   0.064292  6.6214 3.557e-11 ***
## mother_x_large_fam  0.126706   0.089446  1.4166    0.1566
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    649
## Residual Sum of Squares: 606.34
## R-Squared:      0.0719
## Adj. R-Squared: -0.86464
## Chisq: 566.869 on 4 DF, p-value: < 2.22e-16
```

```r
summary(plm(lfs ~ mother + high_skilled + log(nag) + mother_x_large_fam  |
              iv_ssex + high_skilled + mother + log(nag),
            index = c("idmen"),
            data = est_sample))
```

```
## Oneway (individual) effect Within Model
## Instrumental variable estimation
##
## Call:
## plm(formula = lfs ~ mother + high_skilled + log(nag) + mother_x_large_fam |
##     iv_ssex + high_skilled + mother + log(nag), data = est_sample,
##     index = c("idmen"))
##
## Unbalanced Panel: n = 4989, T = 1-2, N = 9940
##
## Residuals:
##     Min.  1st Qu.   Median  3rd Qu.     Max.
## -0.70579 -0.11036  0.00000  0.11036  0.70579
##
## Coefficients:
##                      Estimate Std. Error z-value  Pr(>|z|)
## mother              -0.205149   0.124672 -1.6455  0.099865 .
## high_skilled         0.076276   0.028831  2.6456  0.008154 **
## log(nag)             0.442379   0.103968  4.2550 2.091e-05 ***
## mother_x_large_fam  0.200298   0.367692  0.5447  0.585929
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    649
## Residual Sum of Squares: 629.83
## R-Squared:      0.051434
## Adj. R-Squared: -0.90576
## Chisq: 544.09 on 4 DF, p-value: < 2.22e-16
```

Results on being a mother are significant and negative, as one could expected.

We can run a similar analysis at the **intensive margin**, ie if motherhood and large family impact if the household member works full time or part time:

```r
summary(plm(full_time ~ mother + high_skilled + log(nag) + mother_x_large_fam  |
              iv_twins + high_skilled + mother + log(nag),
            index = c("idmen"),
            data = est_sample))
```

```
## Oneway (individual) effect Within Model
## Instrumental variable estimation
##
## Call:
## plm(formula = full_time ~ mother + high_skilled + log(nag) +
```

```
##     mother_x_large_fam | iv_twins + high_skilled + mother + log(nag),
##     data = est_sample, index = c("idmen"))
##
## Unbalanced Panel: n = 4419, T = 1-2, N = 7118
##
## Residuals:
##       Min.    1st Qu.     Median    3rd Qu.       Max.
## -0.5174579 -0.0018561  0.0000000  0.0018561  0.5174579
##
## Coefficients:
##                       Estimate Std. Error z-value Pr(>|z|)
## mother              -0.0012842  0.0145529 -0.0882   0.9297
## high_skilled         0.0063236  0.0073045  0.8657   0.3866
## log(nag)             0.0444600  0.0343481  1.2944   0.1955
## mother_x_large_fam  0.0362000  0.0474962  0.7622   0.4460
##
## Total Sum of Squares:    43.5
## Residual Sum of Squares: 43.74
## R-Squared:      0.00062142
## Adj. R-Squared: -1.6392
## Chisq: 7.0138 on 4 DF, p-value: 0.13516
```

```
summary(plm(full_time ~ mother + high_skilled + log(nag) + mother_x_large_fam  |
              iv_ssex + high_skilled + mother + log(nag),
            index = c("idmen"),
            data = est_sample))
```

```
## Oneway (individual) effect Within Model
## Instrumental variable estimation
##
## Call:
## plm(formula = full_time ~ mother + high_skilled + log(nag) +
##     mother_x_large_fam | iv_ssex + high_skilled + mother + log(nag),
##     data = est_sample, index = c("idmen"))
##
## Unbalanced Panel: n = 4419, T = 1-2, N = 7118
##
## Residuals:
##     Min.  1st Qu.   Median  3rd Qu.     Max.
## -0.60959 -0.03984  0.00000  0.03984  0.60959
##
## Coefficients:
##                      Estimate Std. Error z-value Pr(>|z|)
## mother              -0.078157   0.065165 -1.1994   0.2304
## high_skilled         0.022114   0.015739  1.4050   0.1600
## log(nag)             0.067713   0.046684  1.4505   0.1469
## mother_x_large_fam  0.297341   0.220712  1.3472   0.1779
##
```

```
## Total Sum of Squares:    43.5
## Residual Sum of Squares: 67.503
## R-Squared:      2.6125e-05
## Adj. R-Squared: -1.6407
## Chisq: 5.98324 on 4 DF, p-value: 0.2004
```

No coefficient can be distinguished from 0. We cannot conclude on an effect of motherhood and family size on labour supply quantity.